

Aware OS

The Next-Generation Operating System for Spatial Computing

License Proprietary

Version 1.0.0

Platform Android | AR Glasses

Overview

Aware OS is a unified, privacy-first operating system developed by **ViiM Technologies** and **AstroTek** that powers the next generation of augmented reality smart glasses and mobile devices. Built on a modified Android Open Source Project (AOSP) base, Aware OS provides seamless cross-device experiences, advanced AR capabilities, and deep integration with the Illumination App ecosystem.

Key Features:

- **Contextual Awareness Engine:** Understands user environment and intent to provide proactive, relevant experiences
- **Advanced AR Rendering:** 6DoF tracking, persistent anchors, spatial audio, and multi-user AR
- **Privacy-First Architecture:** On-device processing, end-to-end encryption, and granular user control
- **Cross-Device Continuity:** Seamless handoff between glasses, phones, and other devices
- **AI Assistant Integration:** Three specialized AI assistants (Gronk AI, Oracle AI, Riley AI)

built into the OS

- **VX Shield Security:** Enterprise-grade security with anti-espionage tools and hardware-backed encryption
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Getting Started

Prerequisites

To develop for Aware OS, you'll need:

- **Aware OS Studio** (based on Android Studio)
- **Java Development Kit (JDK) 17** or higher
- **Android SDK** with API Level 34 or higher
- **ViiM SDK** (included in Aware OS Studio)
- **Physical device** (ViiM Visionaire glasses or vPhone) or emulator

Installation

1. Download Aware OS Studio:

Bash

```
wget https://developer.viimtech.com/downloads/aware-os-studio-latest.tar.gz
tar -xzf aware-os-studio-latest.tar.gz
cd aware-os-studio
./install.sh
```

2. Set up your development environment:

Bash

```
export AWARE_OS_HOME=/path/to/aware-os-studio
export PATH=$PATH:$AWARE_OS_HOME/bin
```

3. Verify installation:

Bash

```
aware-cli --version
# Output: Aware OS CLI v1.0.0
```

Quick Start: Hello AR World

Create your first AR app for Aware OS:

Bash

```
# Create new project
aware-cli create-project --name HelloARWorld --template ar-basic

# Navigate to project directory
cd HelloARWorld

# Build and run on connected device
aware-cli build && aware-cli run
```

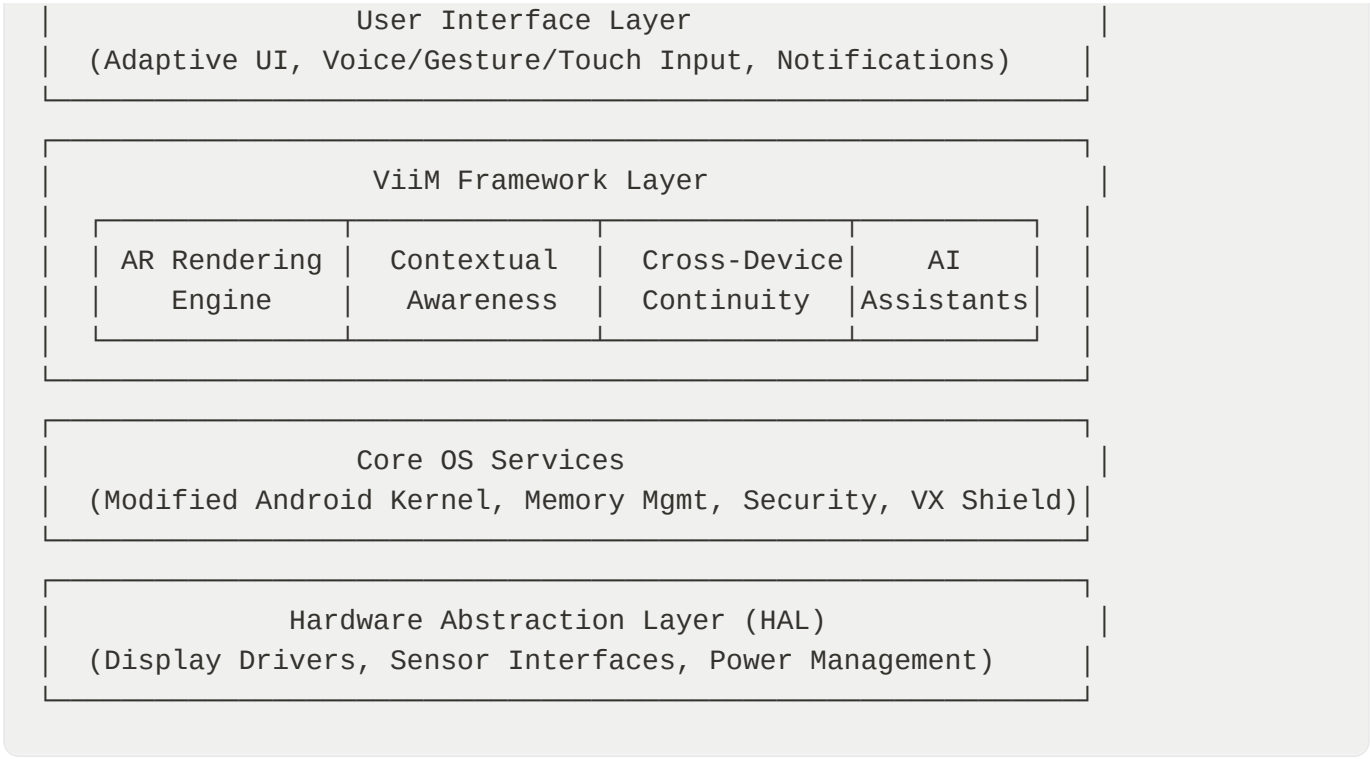
This will create a simple AR app that displays a "Hello World" 3D text object in augmented reality.

Architecture

Aware OS follows a layered architecture designed for performance, security, and extensibility:

Plain Text

```
Application Layer
( Illumination, Digital Diary, Third-Party Apps )
```



Core Components

1. Contextual Awareness Engine

The brain of Aware OS that continuously analyzes user context (location, time, activity, social situation) to provide proactive experiences.

Key Features:

- Automatic mode switching (Morning Rituals, Commute, Work, Explore, Social, Fitness, Nighttime)
- Privacy-preserving on-device processing
- Machine learning-based pattern recognition
- Integration with calendar, location, and sensor data

2. AR Rendering Engine

State-of-the-art augmented reality engine optimized for both glasses and phone displays.

Key Features:

- 6DoF tracking with sub-centimeter accuracy
- Persistent AR anchors (cloud-based and local)
- Environment understanding (plane detection, object recognition, occlusion)
- Spatial audio with head tracking

- Multi-user AR synchronization

3. VX Shield Security Framework

Enterprise-grade security inherited from VX OS, providing comprehensive protection against threats.

Key Features:

- Secure boot and firmware signing
- Hardware-backed encryption (HSM)
- Anti-espionage tools (camera/mic kill switches, network monitoring)
- Privacy dashboard and granular controls
- Secure crypto wallet integration

4. AI Assistant Framework

Three specialized AI assistants built into the OS, accessible from any app:

- **Gronk AI:** Energetic navigator and fitness coach
 - **Oracle AI:** Market predictor and crypto advisor
 - **Riley AI:** Creative companion and content curator
-

Features

For Users

Contextual Intelligence

- Aware OS adapts to your environment and needs automatically
- Proactive suggestions based on location, time, and activity
- Seamless mode switching without manual intervention

Augmented Reality

- Immersive AR experiences on glasses and phones
- Persistent AR content that stays in place across sessions
- Social AR with friends and nearby users

Privacy and Security

- Full control over your data with granular permissions

- On-device processing for sensitive information
- Hardware kill switches for camera and microphone (glasses)
- End-to-end encrypted cloud sync

Cross-Device Continuity

- Start on one device, continue on another seamlessly
- Universal clipboard, notification sync, and app handoff
- Shared photo library and app state across devices

AI Assistance

- Three specialized AI assistants for different needs
- Voice and gesture control for hands-free interaction
- Natural language understanding and contextual responses

For Developers

Comprehensive SDK

- Full Android API compatibility plus ViiM-specific APIs
- AR Development Kit with spatial tracking and rendering
- AI Assistant SDK for integrating Gronk, Oracle, and Riley
- Contextual Awareness SDK for proactive app experiences

Developer Tools

- Aware OS Studio IDE with visual AR scene editor
- Device emulators for glasses and phones
- Performance profiling and debugging tools
- Comprehensive documentation and tutorials

App Distribution

- ViiM App Store with curated AR experiences
- 70/30 revenue share (developer gets 70%)
- In-app purchases and subscription support
- Enterprise app distribution for B2B

For Enterprises

Mobile Device Management (MDM)

- Remote provisioning, configuration, and policy enforcement
- App whitelisting/blacklisting and bulk deployment
- Remote wipe for lost or stolen devices

Enterprise Security

- Single Sign-On (SSO) with corporate identity providers
- Always-on VPN and certificate-based authentication
- Data Loss Prevention (DLP) and clipboard restrictions

Industry Solutions

- Healthcare: HIPAA compliance, hands-free medical records, telemedicine
 - Manufacturing: AR work instructions, remote expert assistance
 - Retail: AR product visualization, virtual try-on, inventory management
 - Education: AR learning experiences, virtual field trips, interactive 3D models
-

Supported Devices

ViiM Visionaire Smart Glasses

Display:

- Waveguide AR display, 40° FOV
- 1280x720 per eye, 2000 nits brightness
- 60 Hz refresh rate

Processor:

- Qualcomm Snapdragon XR2+ Gen 2
- 8GB RAM, 128GB storage
- Dedicated AI/ML accelerator

Sensors:

- Dual 12MP RGB cameras
- Depth sensor (ToF)
- IMU, GPS, ambient light, proximity

Audio:

- Open-ear bone conduction speakers

- 4-mic beamforming array with ANC

Connectivity:

- Wi-Fi 6E, Bluetooth 5.3
- Optional 5G cellular

Battery:

- 500 mAh, 8+ hours typical use
- Fast charging, wireless charging

Physical:

- 50 grams, IP54 water resistance
- Prescription lens compatible

vPhone Devices

Display:

- 6.5" AMOLED, 2400x1080, 120 Hz
- HDR10+, in-display fingerprint sensor

Processor:

- Qualcomm Snapdragon 8 Gen 3
- 12GB RAM, 256GB storage

Cameras:

- Triple rear (50MP main, 12MP ultrawide, 10MP telephoto)
- 32MP front, AR depth sensor

Battery:

- 5000 mAh, 65W fast charging
- 15W wireless charging

Connectivity:

- 5G, Wi-Fi 6E, Bluetooth 5.3
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Developer Documentation

API Reference

Comprehensive API documentation is available at developer.viimtech.com/docs.

Key APIs:

- **AR API:** Spatial tracking, environment understanding, AR rendering
- **Contextual Awareness API:** Access user context, register for mode changes
- **AI Assistant API:** Integrate Gronk, Oracle, and Riley into your apps
- **Digital Locker API:** Crypto wallet, NFT/NFE³ management, payments
- **Cross-Device API:** Implement handoff, sync app state, companion apps

Sample Apps

Explore sample apps in the `/examples` directory:

- **HelloARWorld:** Basic AR object placement
- **TokenHunter:** Location-based AR scavenger hunt
- **ARPhotoGallery:** Persistent AR photo frames
- **FitnessCoach:** Gronk AI integration for workout tracking
- **CryptoPortfolio:** Oracle AI integration for market analysis
- **CreativeStudio:** Riley AI integration for content creation

Tutorials

Step-by-step tutorials are available in the `/docs/tutorials` directory:

1. [Getting Started with Aware OS Development](#)
 2. [Building Your First AR App](#)
 3. [Integrating AI Assistants](#)
 4. [Implementing Cross-Device Continuity](#)
 5. [Publishing to ViiM App Store](#)
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Contributing

We welcome contributions from the community! Please read our [Contributing Guidelines](#) before submitting pull requests.

Development Workflow

1. **Fork the repository**
2. **Create a feature branch:** `git checkout -b feature/amazing-feature`

3. **Commit your changes:** `git commit -m 'Add amazing feature'`
4. **Push to the branch:** `git push origin feature/amazing-feature`
5. **Open a Pull Request**

Code of Conduct

Please note that this project is released with a [Contributor Code of Conduct](#). By participating in this project you agree to abide by its terms.

License

Aware OS is proprietary software owned by ViiM Technologies and AstroTek. The SDK and development tools are available for free to registered developers, but commercial distribution requires a license agreement.

For licensing inquiries, contact: licensing@viimtech.com

Contact

ViiM Technologies

- Website: viimtech.com
- Developer Portal: developer.viimtech.com
- Email: dev@viimtech.com
- Twitter: [@ViiMTech](https://twitter.com/ViiMTech)
- Discord: [ViiM Developer Community](#)

Support:

- Documentation: developer.viimtech.com/docs
 - Community Forums: community.viimtech.com
 - Bug Reports: github.com/viimtech/aware-os/issues
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Roadmap

Q1 2026

- Aware OS 2.0 with enhanced AI capabilities
- Expanded language support (20+ languages)

- Improved battery life (10+ hours for glasses)
- Enhanced gesture recognition accuracy

Q2 2026

- Multi-device AR experiences (glasses + phone + tablet)
- Advanced spatial audio with head tracking
- Developer SDK 2.0 with new AR features
- Enterprise MDM enhancements

2027 and Beyond

- Neural interface support
 - AI-powered real-time translation
 - Advanced health monitoring
 - Quantum-resistant encryption
 - Brain-computer interface (BCI) integration
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Acknowledgments

Aware OS is built on the shoulders of giants. We thank the following open-source projects and communities:

- **Android Open Source Project (AOSP)**
 - **ARCore** (Google)
 - **OpenXR** (Khronos Group)
 - **TensorFlow** and **PyTorch** (AI/ML frameworks)
 - **OpenSSL** (cryptography)
 - And countless other open-source contributors
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Made with ❤️ by Viim Technologies and AstroTek

Empowering the next generation of spatial computing